A FINITE QUASI-COHERENT ALGEBRA IS A DIRECTED COLIMIT OF FINITE FINITELY PRESENTED ALGEBRAS WITH SURJECTIVE TRANSITION MAPS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file properties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the presentation of the algebra as a quotient of the symmetric algebra on a finitely presented module, the approximation of the kernel by finite type submodules giving finitely presented approximants with surjective transition maps, and the monic-relation argument making the approximants finite for large index.

Editorial note. The label lemma-finite-algebra-directed-colimit-finite-finitely-presented names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source properties.tex –label lemma-finite-algebra-directed-colimit-finite-finitely-presented –start 3597 –end 3638 –local-id 28037 returned RESULT=OK (L28037 free, occupancy FREE, public interval clear). Candidate taken from sources/stacks-properties/candidates-halfA.json.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Presentation of a finite quasi-coherent algebra by a finitely presented module. Same chapter, the lemma producing a finitely presented module with a surjection onto the given quasi-coherent module. Supplies the finitely presented module F with a surjection onto the algebra A.
- Description of a quasi-coherent module as a filtered colimit of finite type submodules. Same chapter, the lemma writing a quasi-coherent module as a filtered colimit of its finite type submodules. Supplies the presentation of the kernel of the symmetric algebra surjection as a filtered colimit of finite type submodules.
- Integrality of a finite algebra over an affine open. Algebra, the characterization of a finite algebra over a ring through monic polynomial relations for generators. Supplies the monic polynomials satisfied by the chosen generators over each affine open.

Source reference. Original reference 086M: sources/stacks-properties/pinned-original-source.tex, source lines 3442–3442. Prerequisite pointer invoked at proof line(s) 3608; it names a standard fact of the same chapter that the counted argument

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uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 01PG: sources/stacks-properties/pinned-original-source.tex, source lines 3115–3115. Prerequisite pointer invoked at proof line(s) 3617; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let X be a scheme. Assume X is quasi-compact and quasi-separated. Let $\mathcal{A}$ be a finite quasi-coherent $\mathcal{O}_X$-algebra. Then $\mathcal{A} = \operatorname{colim} \mathcal{A}_i$ is a directed colimit of finite and finitely presented quasi-coherent $\mathcal{O}_X$-algebras such that all transition maps $\mathcal{A}_{i'} \to \mathcal{A}_i$ are surjective.*

Proof. By Lemma 086M there exists a finitely presented $\mathcal{O}_X$-module $\mathcal{F}$ and a surjection $\mathcal{F} \to \mathcal{A}$. Using the algebra structure we obtain a surjection

$$\operatorname{Sym}^*_{\mathcal{O}_X}(\mathcal{F}) \longrightarrow \mathcal{A}$$

Denote $\mathcal{J}$ the kernel. Write $\mathcal{J} = \operatorname{colim} \mathcal{E}_i$ as a filtered colimit of finite type $\mathcal{O}_X$-submodules $\mathcal{E}_i$ (Lemma 01PG). Set

$$\mathcal{A}_i = \operatorname{Sym}^*_{\mathcal{O}_X}(\mathcal{F})/(\mathcal{E}_i)$$

where $(\mathcal{E}_i)$ indicates the ideal sheaf generated by the image of $\mathcal{E}_i \to \operatorname{Sym}^*_{\mathcal{O}_X}(\mathcal{F})$. Then each $\mathcal{A}_i$ is a finitely presented $\mathcal{O}_X$-algebra, the transition maps are surjections, and $\mathcal{A} = \operatorname{colim} \mathcal{A}_i$. To finish the proof we still have to show that $\mathcal{A}_i$ is a finite $\mathcal{O}_X$-algebra for i sufficiently large. To do this we choose an affine open covering $X = U_1 \cup \ldots \cup U_m$. Take generators $f_{j,1}, \ldots, f_{j,N_j} \in \Gamma(U_i, \mathcal{F})$. As $\mathcal{A}(U_j)$ is a finite $\mathcal{O}_X(U_j)$-algebra we see that for each k there exists a monic polynomial $P_{j,k} \in \mathcal{O}(U_j)[T]$ such that $P_{j,k}(f_{j,k})$ is zero in $\mathcal{A}(U_j)$. Since $\mathcal{A} = \operatorname{colim} \mathcal{A}_i$ by construction, we have $P_{j,k}(f_{j,k}) = 0$ in $\mathcal{A}_i(U_j)$ for all sufficiently large i. For such i the algebras $\mathcal{A}_i$ are finite. □